ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme	:	Bachelor of Information Technology in Internet of Things
Course Code & Name	:	IIB43203 Cloud Computing
Course Lecturer(s)	:	AP Dr Megat NorulAzmi
Submission Title	:	Group Project
Deadline	:	Day 7 Month June Year 2026 Time 11.59pm
Penalties	:	• 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted. • Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (√) each box below if you agree:

- √ I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
- √ This submission is my own, unless indicated with proper referencing.
- √ This submission has not been previously submitted or published.
- √ This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
	SYAHMI BIN MUHAMMAD SHAMSUL HASHIR MALIK	52224224168 52227124933

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
	IIB43203	Create a Web Application using AWS services	52224224168 52227124933

Universiti Kuala Lumpur
Malaysian Institute of Information Technology

Title:
Create a Web Application using AWS Services

Course:
IIB43203 Cloud Computing

Semester:
March 2026

Prepared by:
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Prepared for:
AP Dr Megat NorulAzmi

1. Introduction

This project was developed to demonstrate how cloud computing services can be used to build and deploy a modern web application. By leveraging Amazon Web Services (AWS), we aimed to integrate Compute, Storage, Networking, and Database solutions into a single functional system. AWS was chosen because it is the leading cloud platform, offering scalability, reliability, and a wide range of services that are accessible through its free trial account. The objective of this project is not only to create a working application but also to provide a comprehensive instructional guide that allows others to replicate the process.

The scope of this project covers the design, deployment, and documentation of a web application using four core AWS service categories:

- **Storage (Amazon S3):** Manages and stores files, media, and backups in a secure and scalable environment.

- **Networking (Amazon VPC, Security Groups):** Ensures secure communication between services and controls how data flows in and out of the application.

- **Compute (Amazon EC2):** Provides the processing power to run the application and host the web server.

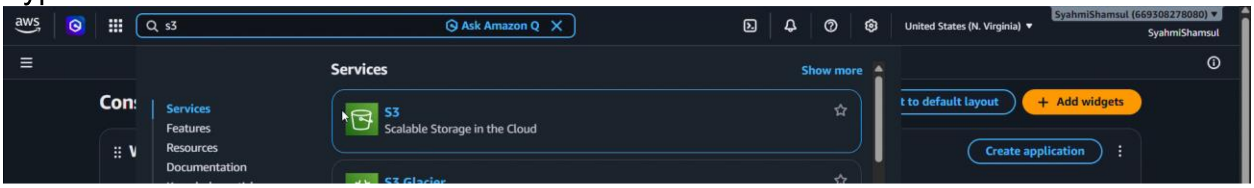
- **Database (Amazon RDS):** Organizes and manages structured data, enabling efficient storage and retrieval for the application.

This project also includes a step-by-step tutorial guide, a video presentation where all team members contribute, and a GitHub repository containing all project materials. Together, these deliverables showcase teamwork, creativity, and practical cloud computing skills.

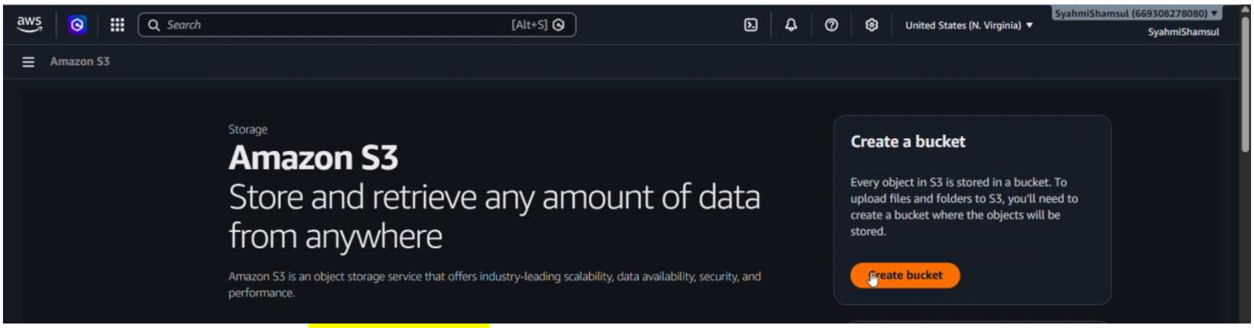
2. Storage (Amazon S3)

Open your AWS console.

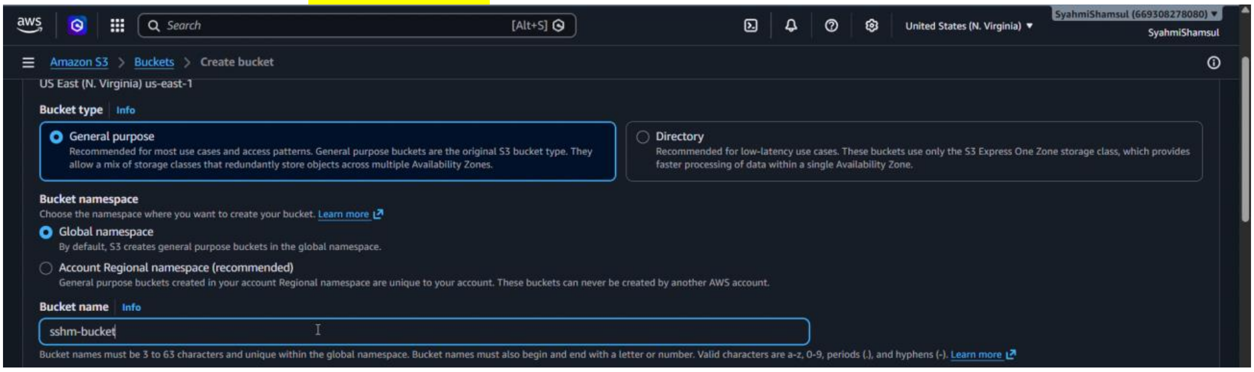
Type “S3” in the search bar.



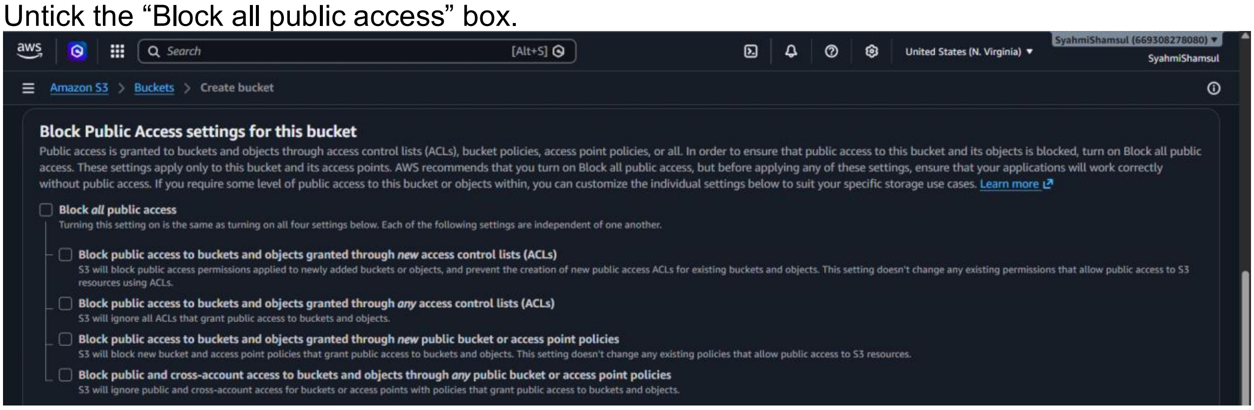
Click “Create bucket” button



Enter bucket name = sshm-bucket

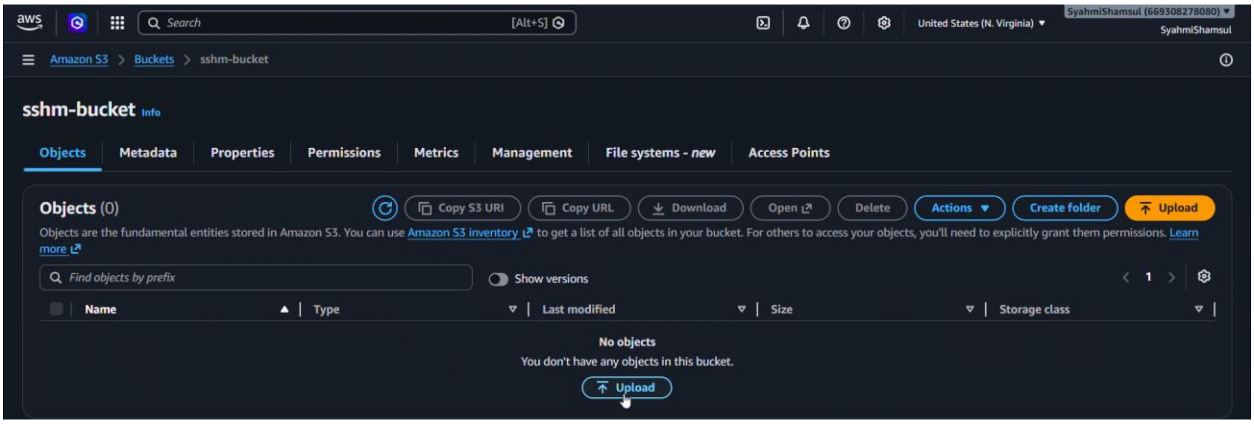


Untick the “Block all public access” box.

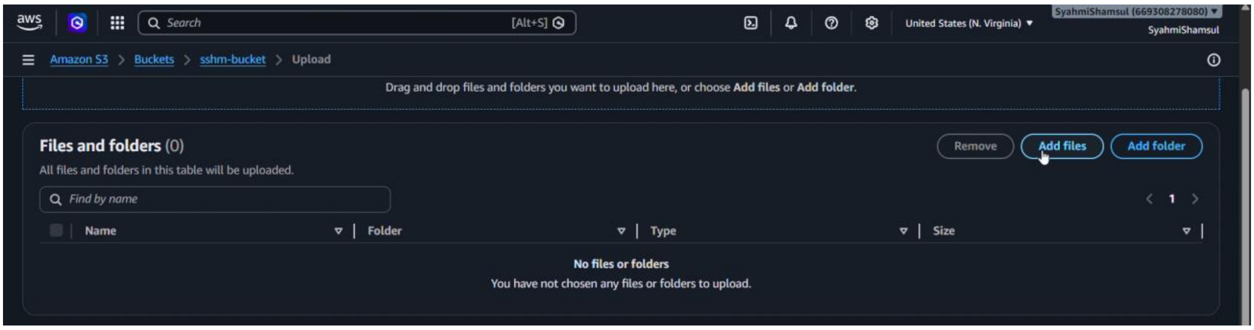


After finished setting up, click the “Create bucket” button at the bottom of the page.

After bucket has been created, click “Upload” button.



Click “add file” button.

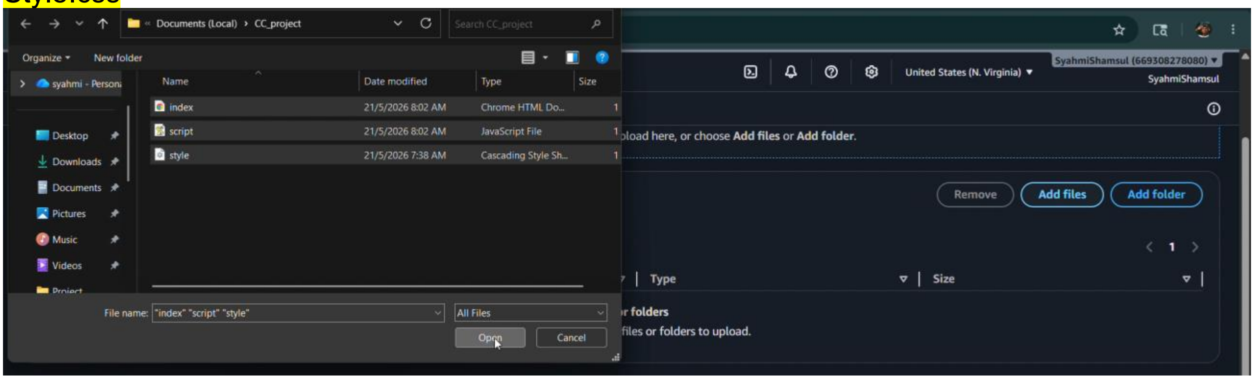


Add files from your computer.

Index.html

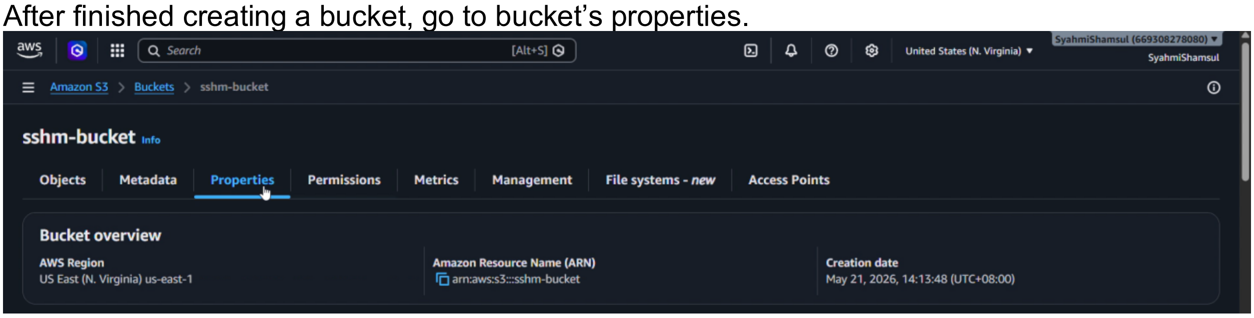
Script.js

Style.css



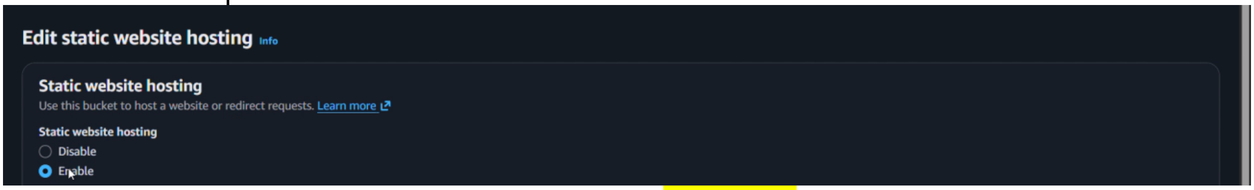
Click “Upload” button at the bottom of the page.

After finished creating a bucket, go to bucket’s properties.

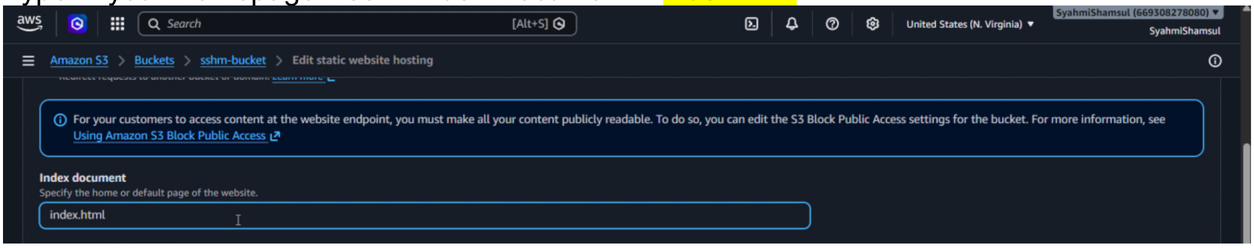


Scroll to the Static Website Hosting section and click “Edit”.

Select “enable” option

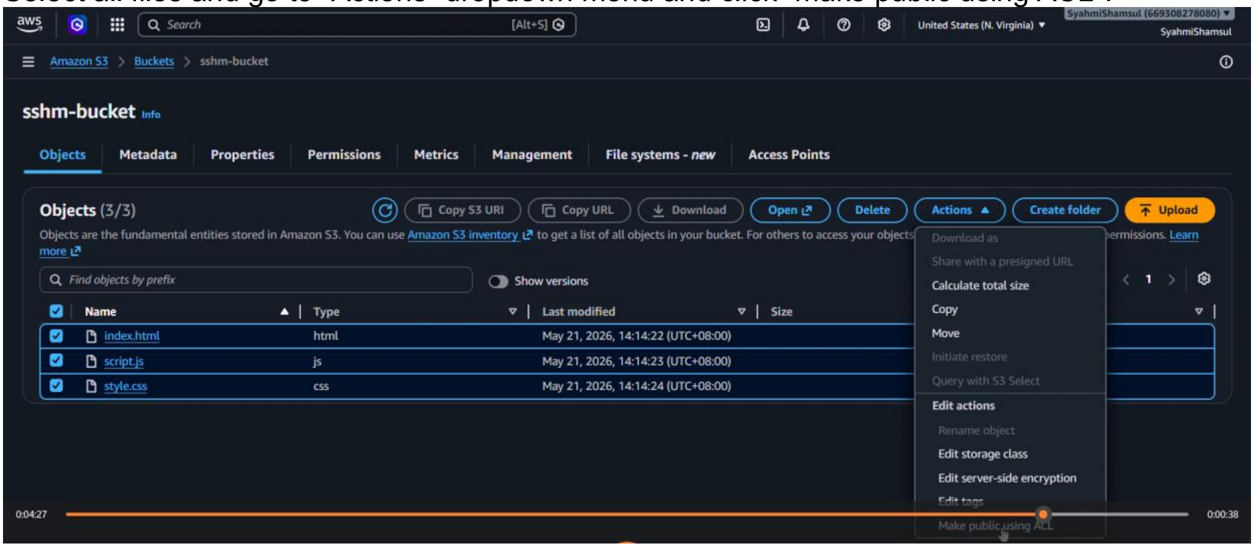


Type in your Homepage files in Index Document = index.html

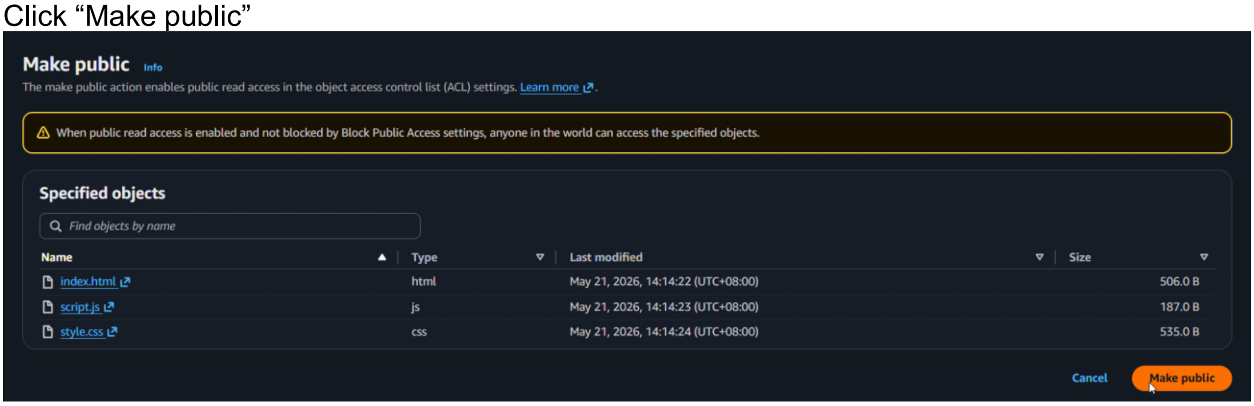


After finished setting up, click “Save changes” button at the bottom of the page. Take note of the website address.

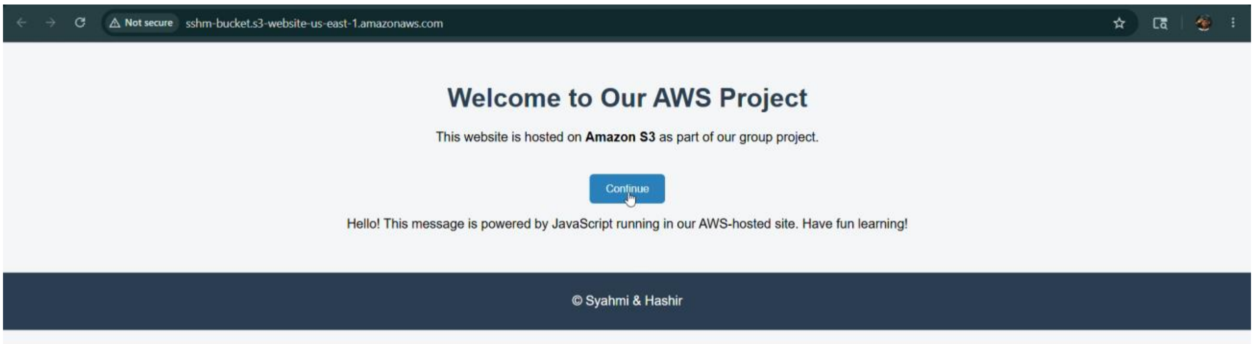
Select all files and go to “Actions” dropdown menu and click “make public using ACL”.



Click “Make public”



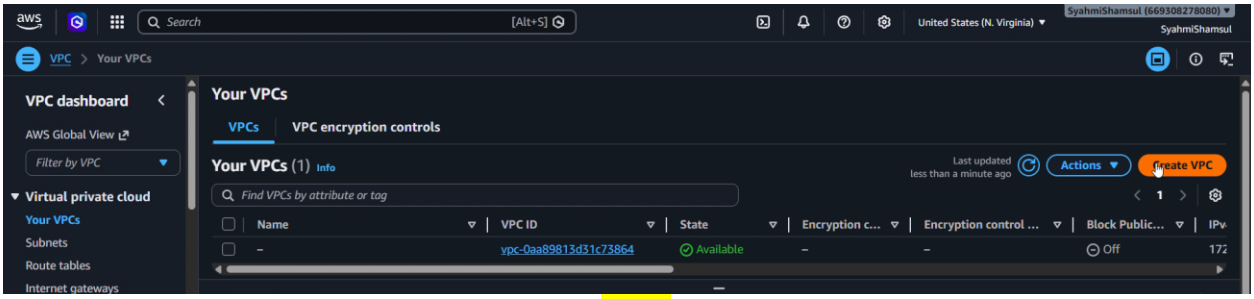
Open the website address


Not secure
sshm-bucket.s3-website-us-east-1.amazonaws.com
Welcome to Our AWS Project
This website is hosted on Amazon S3 as part of our group project.
Continue
Hello! This message is powered by JavaScript running in our AWS-hosted site. Have fun learning!
© Syahmi & Hashir

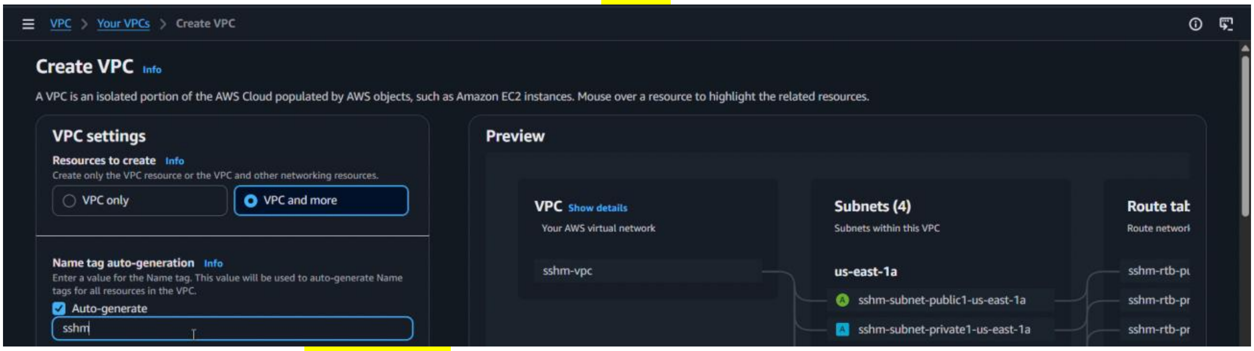
3. Networking (Amazon VPC)

Type VPC in the search bar

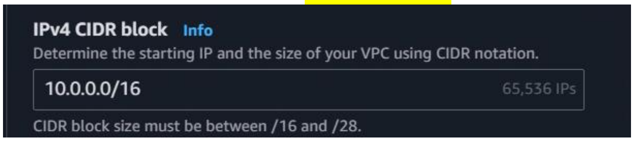
Click “Create VPC” Button



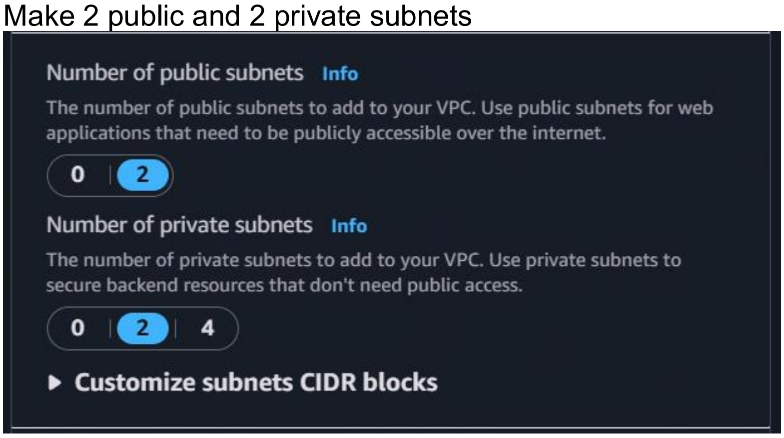
Select VPC and more and enter VPC name = sshm



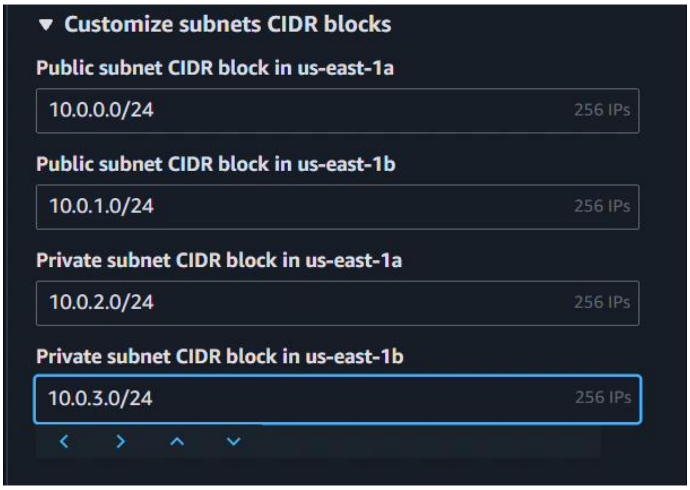
Set IPv4 CIDR block = 10.0.0.0/16



Make 2 public and 2 private subnets

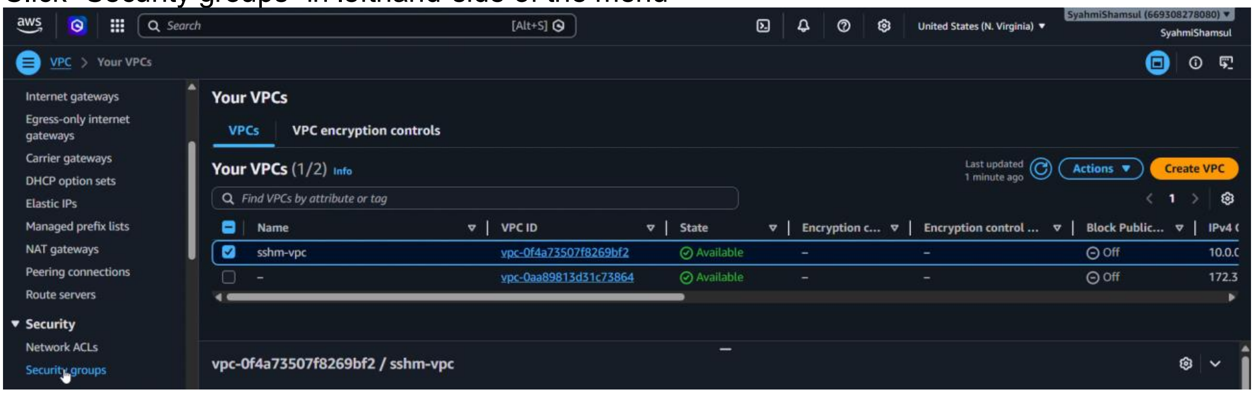


Inside Customize CIDR block menu, set each subnet CIDR block:
Public-1a = 10.0.0.0/24
Public-1b = 10.0.1.0/24
Private-1a = 10.0.2.0/24
Private-1b =10.0.3.0/24

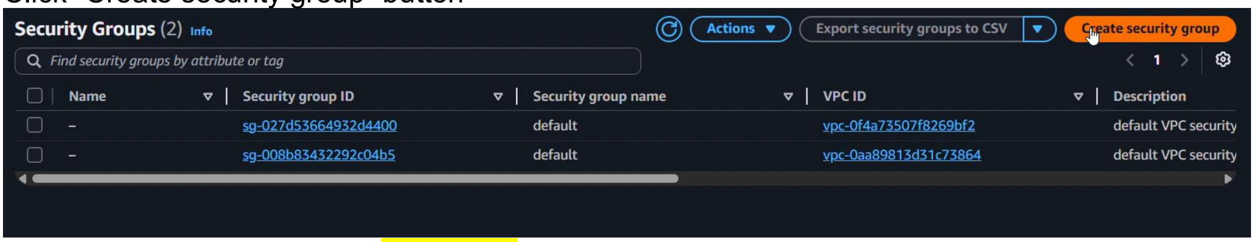


After finished setting up, click “Create VPC” button at the bottom of the page.

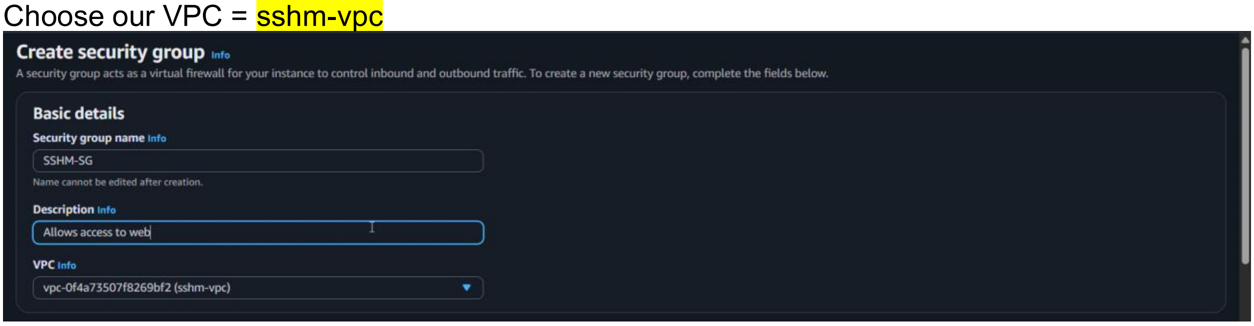
Click “Security groups” in lefthand-side of the menu



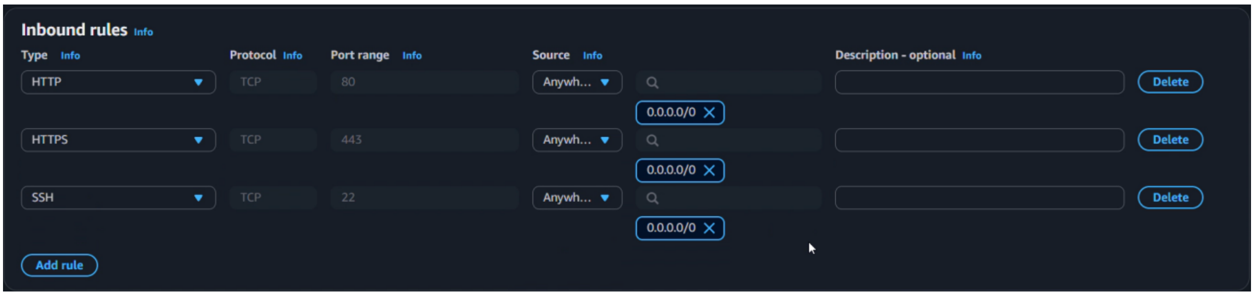
Click “Create security group” button



Enter security group name = SSHM-SG
Enter security description
Choose our VPC = sshm-vpc

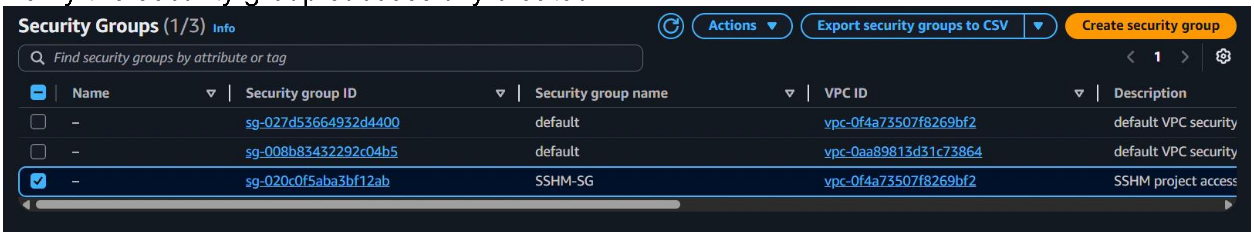


In Inbound rules section, add rule of HTTP, HTTPS and SSH with source of Anywhere-IPv4 (0.0.0.0/0).



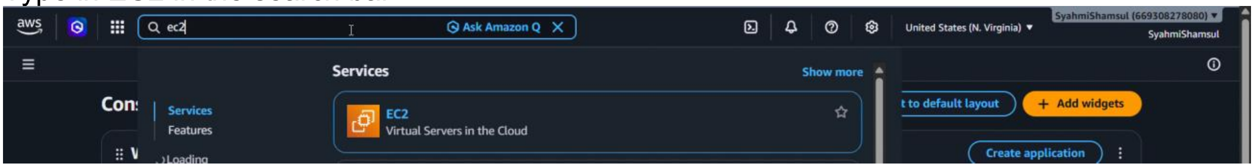
After finished setting up, click “Create security group” button at the bottom of the page.

Verify the security group successfully created.

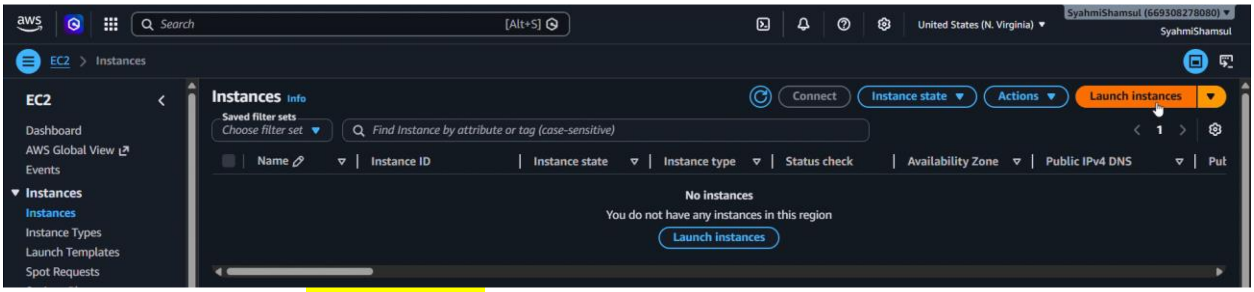


4. <u>Computing (Amazon EC2)</u>

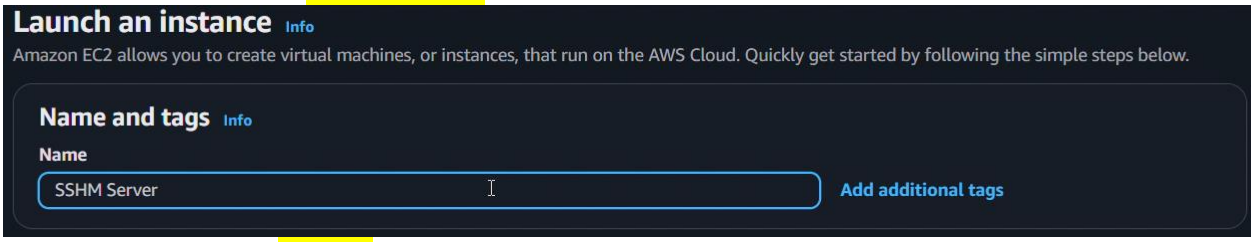
Type in EC2 in the search bar



Click “Launch instances” button

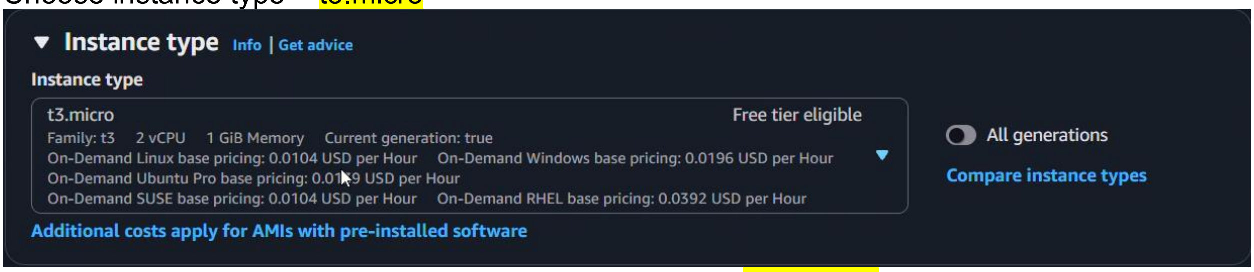


Enter instance name = SSHM Server

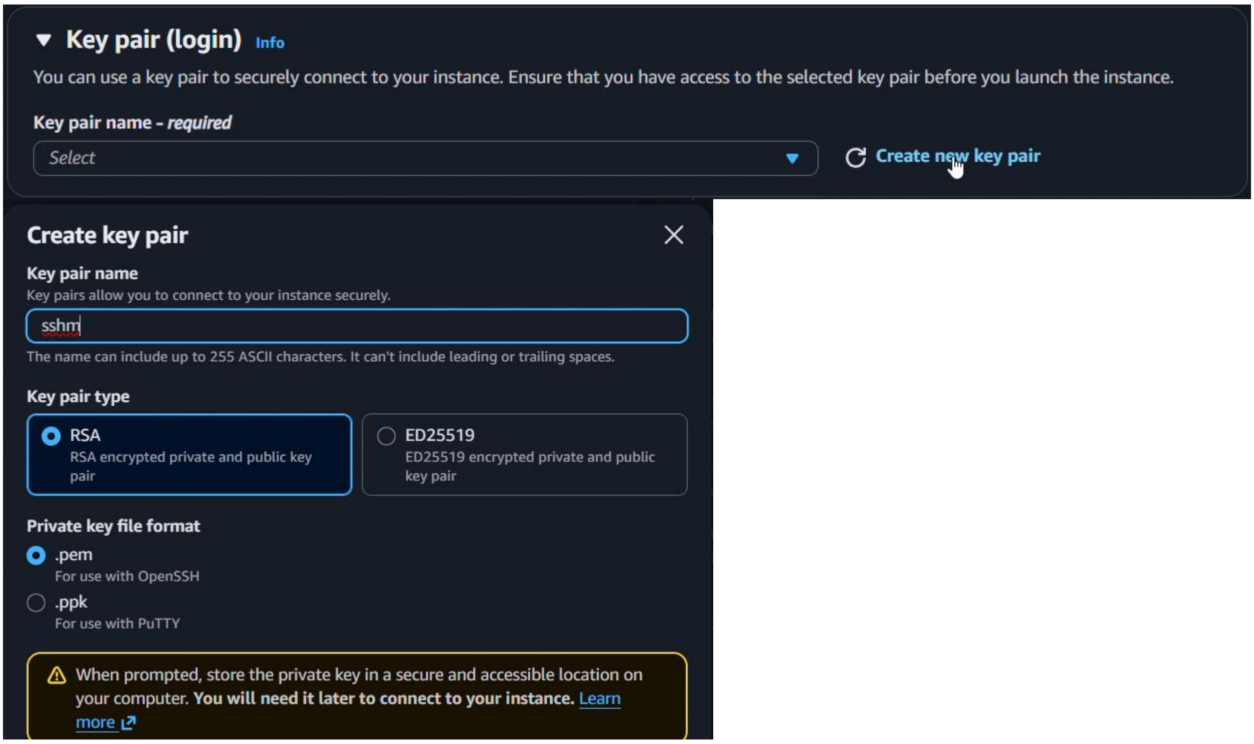


Choose OS Image = Ubuntu

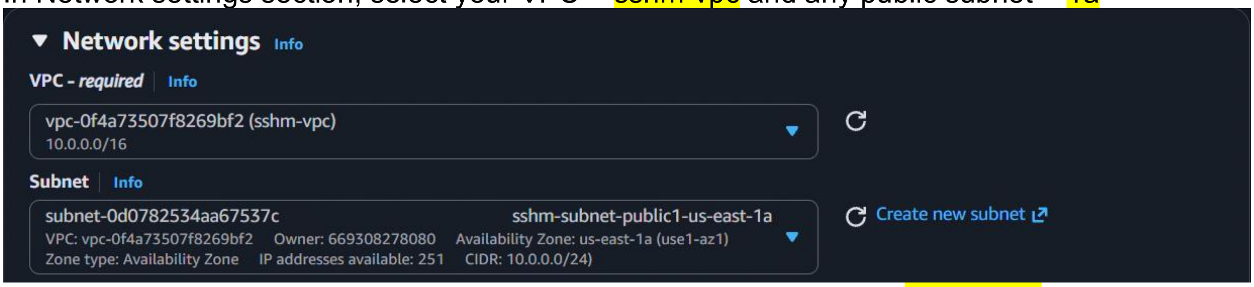
Choose instance type = t3.micro



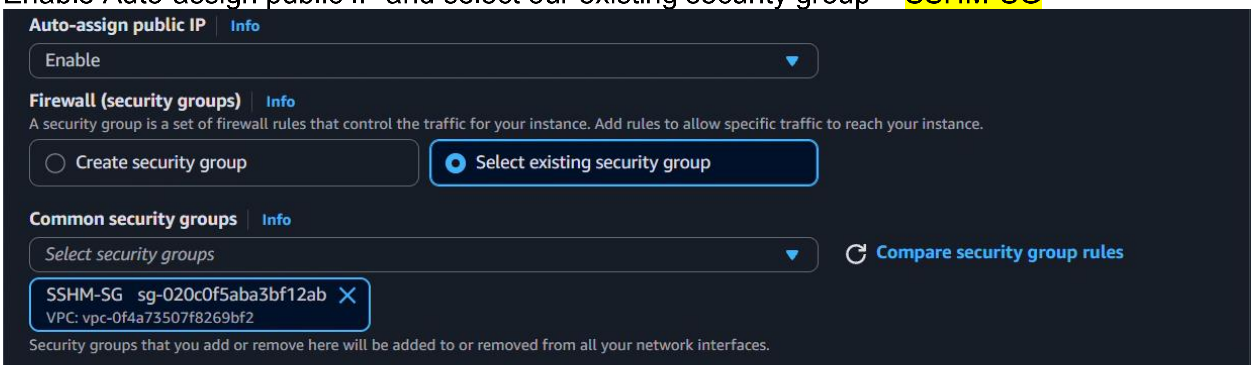
Create new key pair for login process into the instance = sshm.pem



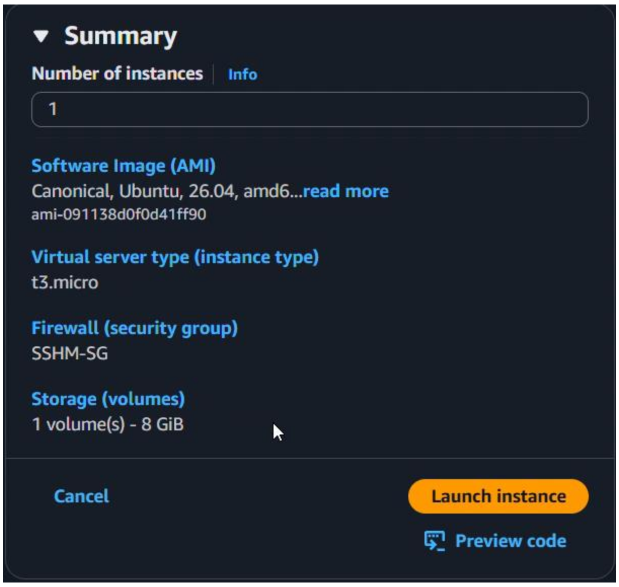
In Network settings section, select your VPC = sshm-vpc and any public subnet = 1a



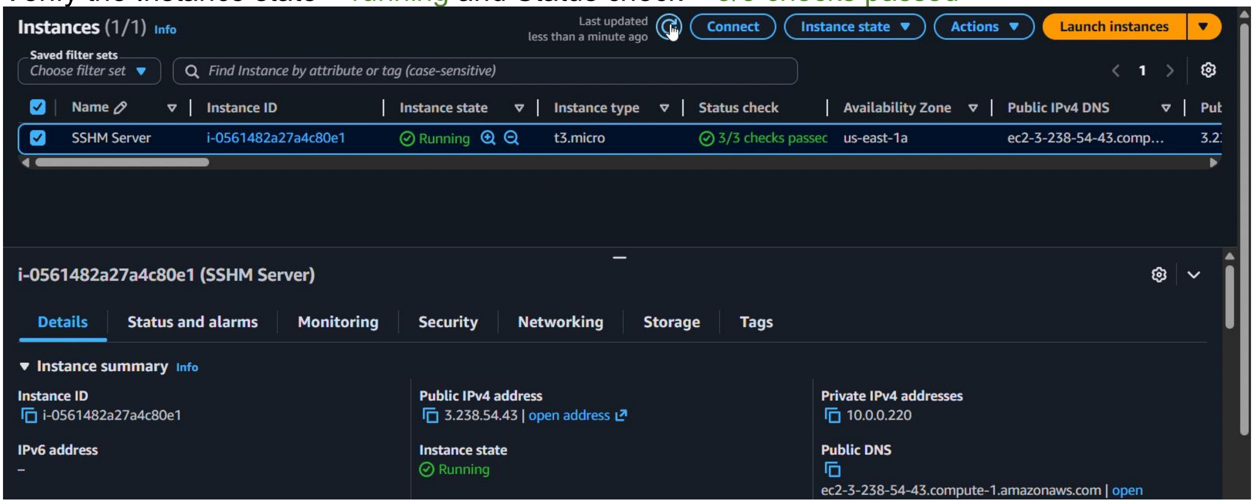
Enable Auto-assign public IP and select our existing security group = SSHM-SG



In the Summary section at the righthand-side, click “Launch Instance” button.

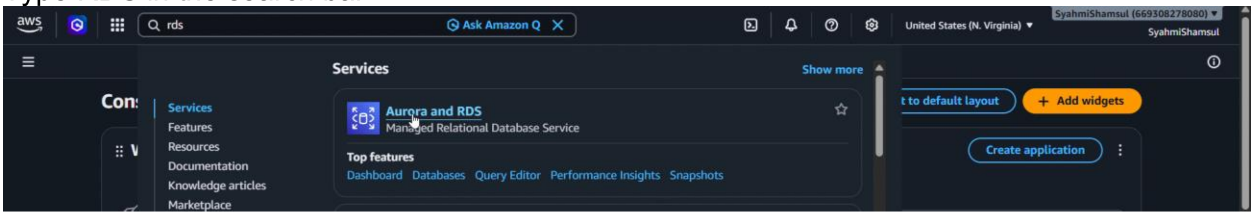


Verify the Instance state = running and Status check = 3/3 checks passed

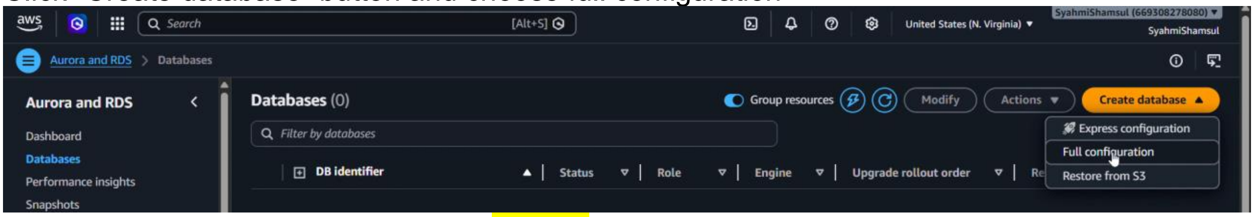


5. Database (Amazon RDS)

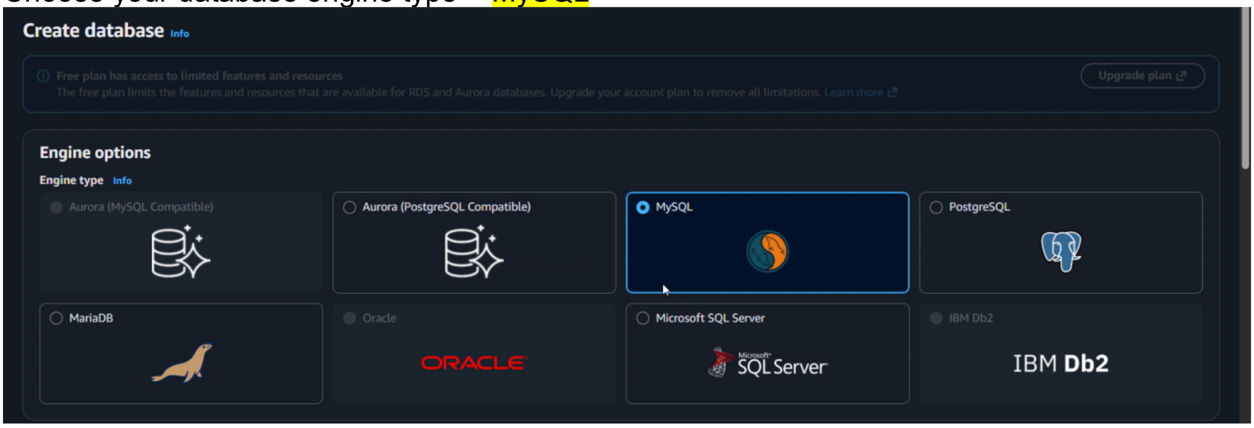
Type RDS in the search bar



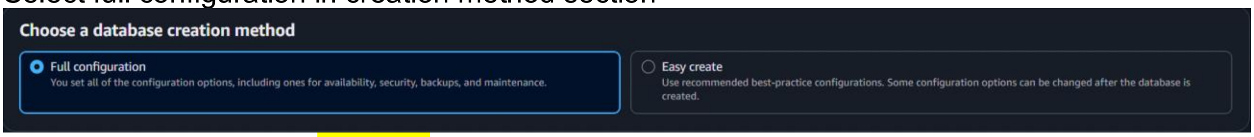
Click “Create database” button and choose full configuration



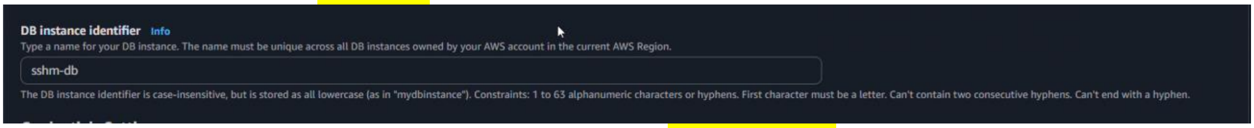
Choose your database engine type = MySQL



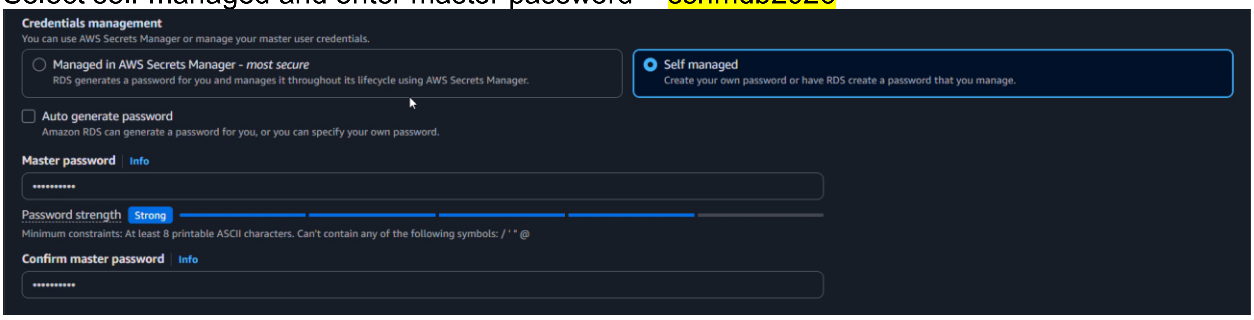
Select full configuration in creation method section



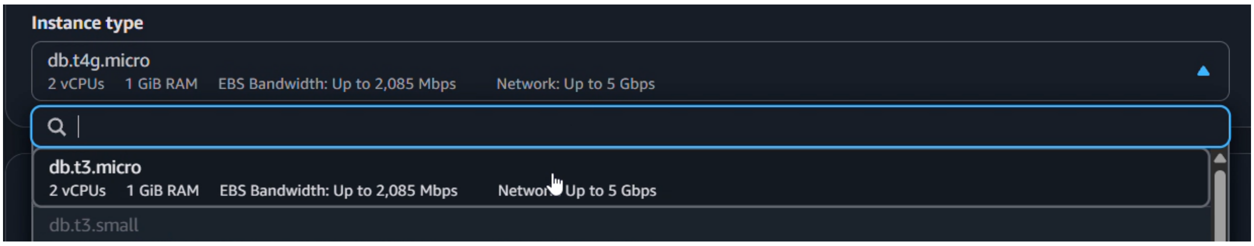
Enter database name = sshm-db



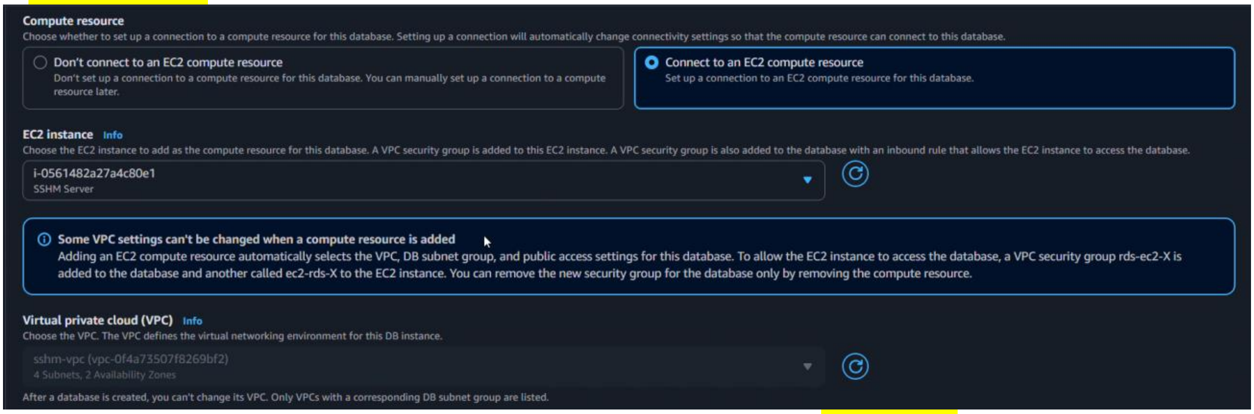
Select self-managed and enter master password = sshmdb2026



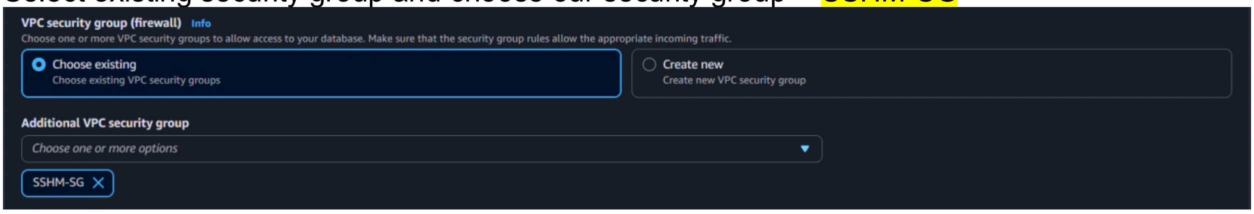
Set instance type = t3.micro



In connectivity section, select connect to EC2 compute resource. Then select our EC2 instance = SSHM Server

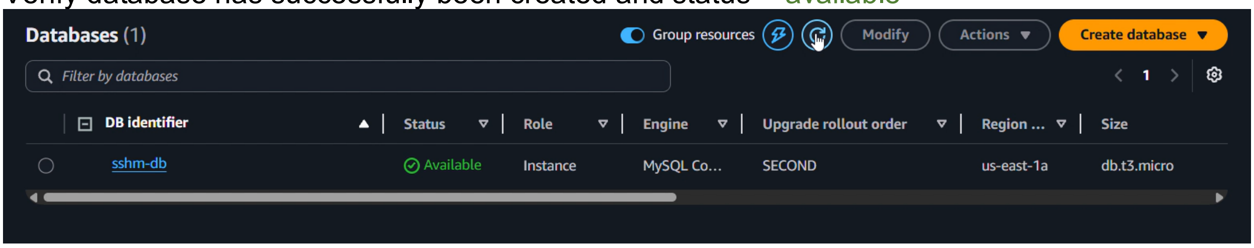


Select existing security group and choose our security group = SSHM-SG



After finished setting up, click “Create database” button at the bottom of the page.

Verify database has successfully been created and status = available



6. How to Run Application

Open terminal in local machine

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> s
```

Prompt *ssh –i <.pem directory> ubuntu@<EC2 public IP>* to login into your EC2 instance

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> ssh -i C:\Users\cools\Downloads\sshm.pem ubuntu@54.164.204.115
```

Output:

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu May 21 09:26:26 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   35.4% of 6.61GB   Processes:             122
  Memory usage: 25%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 10.0.0.220

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

30 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu May 21 09:19:29 2026 from 115.135.30.6
ubuntu@ip-10-0-0-220:~$
```

Prompt *sudo apt install apache2 php php-mysql* to install apache inside EC2 instance.

```
ubuntu@ip-10-0-0-220:~$ sudo apt install apache2 php php-mysql
```

Output:

```
apache2 is already the newest version (2.4.66-2ubuntu2.1).
php is already the newest version (2:8.5+99ubuntu1).
php-mysql is already the newest version (2:8.5+99ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 30
```

Prompt *sudo apt install mysql-client* to install mysql client inside EC2 instance.

```
ubuntu@ip-10-0-0-220:~$ sudo apt install mysql-client
```

Output:

```
Installing:
  mysql-client

Installing dependencies:
  mysql-common

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 30
  Download size: 16.1 kB
  Space needed: 88.1 kB / 4570 MB available

Continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 mysql-common all 5.8+1.1.1ubuntu2 [7002 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 mysql-client amd64 8.4.8-0ubuntu1 [9136 B]
Fetched 16.1 kB in 0s (826 kB/s)
Selecting previously unselected package mysql-common.
(Reading database ... 86208 files and directories currently installed.)
Preparing to unpack .../mysql-common_5.8+1.1.1ubuntu2_all.deb ...
Unpacking mysql-common (5.8+1.1.1ubuntu2) ...
Selecting previously unselected package mysql-client.
Preparing to unpack .../mysql-client_8.4.8-0ubuntu1_amd64v3.deb ...
Unpacking mysql-client (8.4.8-0ubuntu1) ...
Setting up mysql-common (5.8+1.1.1ubuntu2) ...
update-alternatives: using /etc/mysql/my.cnf.fallback to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Setting up mysql-client (8.4.8-0ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

Prompt *sudo nano /var/www/html/<file name>* to create new file inside EC2 acting as a website ui and database connection.

```
ubuntu@ip-10-0-0-220:~$ sudo nano /var/www/html/index2.php
```

Output:

```
  GNU nano 8.7.1                                                    /var/www/html/index2.php *
            border-radius: 6px;
            cursor: pointer;
            transition: background-color 0.3s ease;
        }
        button:hover {
            background-color: #45a049;
        }
    </style>
</head>
<body>
    <h1>SSHM Student Information</h1>
    <p>Click continue to see student database</p>

    <button onclick="window.location.href='sshm_db.php'">
        continue
    </button>
</body>
</html>
```

```
ubuntu@ip-10-0-0-220:~$ sudo nano /var/www/html/sshm_db.php
```

Output:

```
<?php
error_reporting(E_ALL);
ini_set('display_errors', 1);

$servername = "sshm-db.cerwuoiqwew3.us-east-1.rds.amazonaws.com";
$username   = "admin";   // master user
$password   = "sshmdb2026";   // master password
$dbname     = "sshm_db";         // database you created

$conn = mysqli_connect($servername, $username, $password, $dbname);

if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

// Query the students table
$sql = "SELECT id, name, course, year FROM student";
$result = mysqli_query($conn, $sql);

if ($result) {
    echo "<h2>✅ Connected successfully!</h2>";
    echo "<table border='1' cellpadding='5'>";
    echo "<tr><th>ID</th><th>Name</th><th>Course</th></tr>";
    while ($row = mysqli_fetch_assoc($result)) {
        echo "<tr><td>".$row['id']."</td><td>".$row['name']."</td><td>".$row['course']."</td></tr>".$row['year']."</td></tr>";
    }
    echo "</table>";
} else {
    echo "❌ Query failed: " . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

Prompt *sudo systemctl start apache2* to restart apache

```
ubuntu@ip-10-0-0-220:~$ sudo systemctl start apache2
```

Prompt *sudo systemctl enable apache2* to enable apache

```
ubuntu@ip-10-0-0-220:~$ sudo systemctl enable apache2
```

Output:

```
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

Prompt *mysql –h <database end point> -p 3306 –u <master username>* -p

```
ubuntu@ip-10-0-0-220:~$ mysql -h sshm-db.cerwuoiqwew3.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
```

Enter database password

```
Enter password:
```

Output:

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 122
Server version: 8.4.8 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```
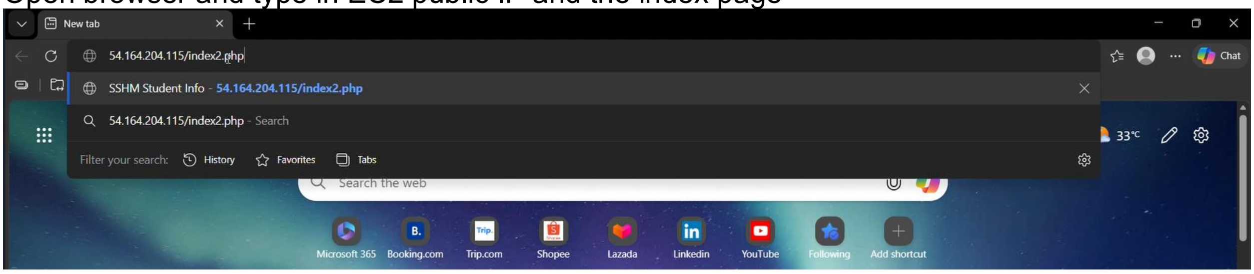

Enter MySQL code

```
mysql> CREATE DATABASE sshm_db;
Query OK, 1 row affected (0.05 sec)

mysql> USE sshm_db;
EATE TABLE student (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    course VARCHAR(50) NOT NULL,
    year INT NOT NULL
);

INSERT INTO student (name, course, year) VALUES
('Syahmi', 'IoT', 3),
('Hasyir', 'IoT', 2),
('Farah', 'AI', 4);

SELECT * FROM student;Database changed
mysql>
mysql> CREATE TABLE student (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(50) NOT NULL,
    ->     course VARCHAR(50) NOT NULL,
    ->     year INT NOT NULL
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO student (name, course, year) VALUES
    -> ('Syahmi', 'IoT', 3),
    -> ('Hasyir', 'IoT', 2),
    -> ('Farah', 'AI', 4);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
```

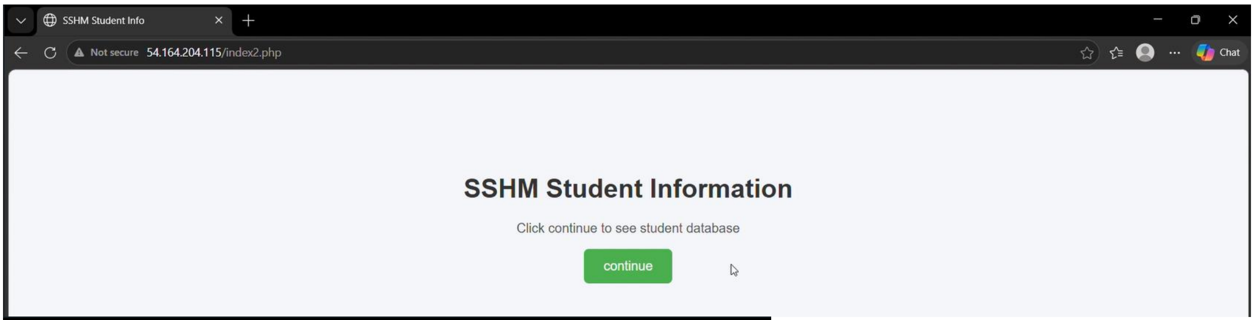
Output:

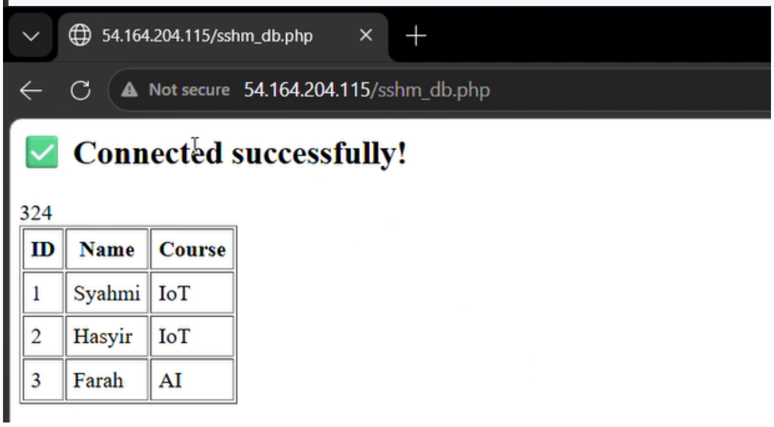
```
mysql> SELECT * FROM student;
+----+--------+--------+------+
| id | name   | course | year |
+----+--------+--------+------+
|  1 | Syahmi | IoT    |    3 |
|  2 | Hasyir | IoT    |    2 |
|  3 | Farah  | AI     |    4 |
+----+--------+--------+------+
3 rows in set (0.00 sec)
```

Open browser and type in EC2 public IP and the index page



Output:


SSHM Student Info
Not secure 54.164.204.115/index2.php
Chat
SSHM Student Information
Click continue to see student database
continue


54.164.204.115/sshm_db.php
Not secure 54.164.204.115/sshm_db.php
Connected successfully!
324
ID Name Course
1 Syahmi IoT
2 Hasyir IoT
3 Farah AI

7. Conclusion

In this project, we successfully designed and deployed a web application using four core AWS services: Compute, Storage, Networking, and Database. Each service played a critical role in ensuring that the application was functional, secure, and scalable. Compute allowed us to run the application on virtual servers, Storage provided a reliable way to manage files and data, Networking ensured secure communication between components, and Database organized and retrieved structured information efficiently.

Through this project, we gained valuable hands-on experience with AWS, learned how different services integrate to form a complete cloud solution, and strengthened our teamwork skills by dividing responsibilities among group members. The deliverables — including the step-by-step guide, video presentation, and GitHub repository — demonstrate not only our technical understanding but also our ability to communicate and document our work clearly.

Overall, this project highlights the importance of cloud computing in modern application development and shows how AWS empowers users to build reliable, scalable, and innovative solutions.

GitHub repository:

https://github.com/MikazeMimi/CloudComputing_GroupProject_March2026.git

Youtube link:

https://youtu.be/27LMjuUfeMc